

ML Engineering Field Guide

A practical competency guide for production machine-learning engineering.

How to use this guide

Use these topics as a study and evidence checklist. Competra does not treat reading a guide as proof of competency; pair learning with practice, a project task and an unfamiliar-context assessment.

1. Python engineering and packaging

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

2. Data validation and SQL

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

3. Model evaluation and reproducibility

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

4. API serving with FastAPI

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

5. Testing and observability

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

6. Docker and deployment

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.

7. Technical defense and trade-offs

Evidence target: explain the concept, apply it in a small task, test the behavior, and be able to defend one design decision. Suggested proof should be observable in a project, challenge or assessment rather than inferred from completion alone.